

Technology Stack

Date	5 July 2026
Team ID	SWTID-2026-2638
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular; HTML/CSS)	React.js (with Vite), HTML5, CSS3, JavaScript (ES6+)	React.js provides a fast and responsive UI with reusable components. Vite offers a faster build and development experience. HTML5, CSS3, and JavaScript ensure modern, responsive, and cross-browser compatible UI.
2	Backend / Server-Side (e.g., Node.js, Python; Spring Boot)	Python, FastAPI	FastAPI is a modern, high-performance Python web framework ideal for building RESTful APIs with automatic validation and documentation.
3	Database (e.g., MySQL, PostgreSQL; MongoDB)	SQLite	SQLite is lightweight, serverless, and easy to set up. It is ideal for this application's data storage needs and ensures portability.
4	ORM / Database Toolkit (e.g., SQLAlchemy, Prisma)	SQLAlchemy ORM	SQLAlchemy ORM provides powerful and flexible database interaction with Python, making it easier to manage models and queries.
5	AI / ML Services (e.g., OpenAI API, Google Gemini API)	Google Gemini AI API	Gemini AI API is used for financial analysis, settlement prediction, negotiation strategies, and AI letter generation.
6	API Communication (e.g., REST, GraphQL)	RESTful APIs	RESTful APIs are used for communication between frontend and backend, ensuring scalability and easy integration.
7	Development Tools (e.g., VS Code, Git)	VS Code, Git, GitHub	VS Code is used as the IDE. Git and GitHub are used for version control, collaboration, and code management.
8	Deployment (e.g., Vercel, Render, AWS, Docker)	Render (Backend), Vercel (Frontend)	Render is used to deploy the FastAPI backend, and Vercel is used for deploying the React frontend for fast and reliable hosting.